

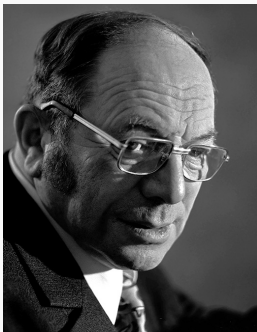
Fundamentals of the Monge–Kantorovich Theory

Optimal Transport and Stochastic Analysis · Lecture 1 of 3

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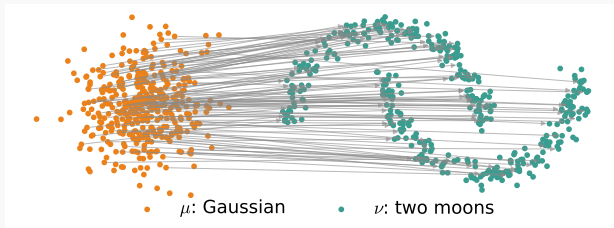


Leonid Kantorovich, Nobel Memorial Prize in Economics, 1975

“Soon after this work began it became clear that the methods being developed there could find a far larger field of application . . .”

— L. Kantorovich, *The Best Use of Economic Resources*, 1965

Transport overview



- ▶ Move a probability μ onto another ν with least total cost, say quadratic cost $\|y - x\|^2$.
- ▶ The answer is a **map** T with $T_{\#}\mu = \nu$ (left: 400 points, exactly solved).
- ▶ Generative modelling asks for the same object: a push-forward of noise onto data.
- ▶ Every few decades OT is re-energised by a new field: economics, PDE, geometry, probability, statistics, ML.

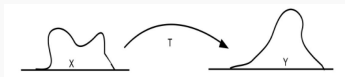
The fundamentals: plan for today

- ▶ Brenier's theorem identifies optimal transport maps as **gradients of convex functions**.
- ▶ This lets us draw “straight lines” between two probability measures: McCann / displacement interpolations, i.e. **geodesics**.
- ▶ Functionals convex along these lines are **geodesically convex**; they matter in applications.

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- 1 Monge and Kantorovich; **Kantorovich duality**
 - 2 Convex functions; **Brenier's theorem**; the twist condition
 - 3 **McCann interpolation**
 - 4 Dynamic OT: **Benamou–Brenier** and the path space
 - 5 Wasserstein space and geodesic convexity
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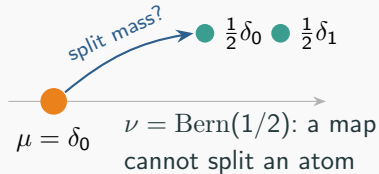
Monge and Kantorovich

The Monge problem (1781)



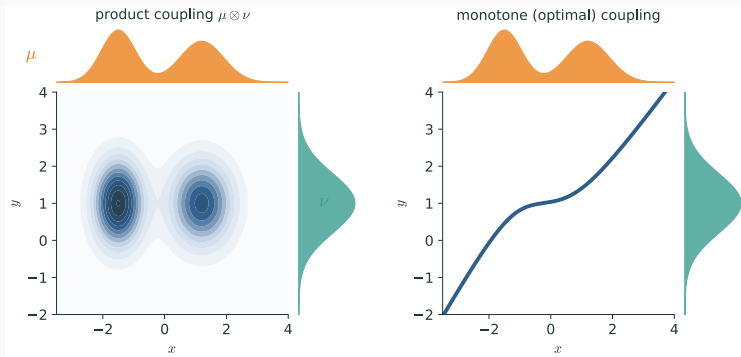
Gaspard Monge, *Déblais et remblais*

- ▶ μ, ν probabilities on $\mathcal{X} = \mathbb{R}^d = \mathcal{Y}$ with finite second moments.
- ▶ Minimize $\mathbb{E} \|T(X) - X\|$ over maps $T : \mathbb{R}^d \rightarrow \mathbb{R}^d$ such that $X \sim \mu \Rightarrow T(X) \sim \nu$.
- ▶ Such a T may not even exist!



Kantorovich's relaxation: couplings

- ▶ A **coupling** of (μ, ν) is a joint law on $\mathbb{R}^d \times \mathbb{R}^d$ with marginals μ and ν .
- ▶ $\Pi(\mu, \nu)$ is the set of couplings.
- ▶ If T exists, the law of $(X, T(X))$ is a coupling. Couplings may split mass.



The Monge–Kantorovich problem

- ▶ (Kantorovich 1942) Let $c(x, y)$ - cost of transporting from x to y . Minimize over $\Pi(\mu, \nu)$:

$$\inf_{\gamma \in \Pi(\mu, \nu)} \mathbb{E}_{\gamma}[c(X, Y)], \quad \text{e.g.} \quad c(x, y) = \|y - x\|^2.$$

- ▶ **Linear** in γ ; $\Pi(\mu, \nu)$ is convex. An infinite-dimensional linear program.
- ▶ Lower semicontinuity + weak compactness of $\Pi(\mu, \nu) \Rightarrow$ an optimal coupling exists.
- ▶ What does the optimal coupling look like? Is it a Monge map?

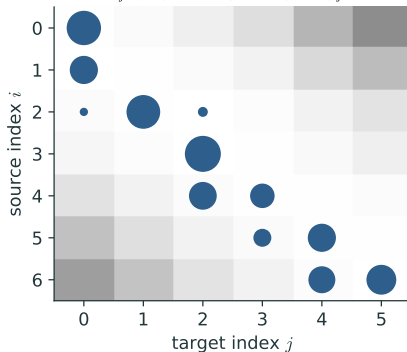
Kantorovich (1939, 1942); Koopmans; Dantzig's simplex method (1947). Nobel 1975: Kantorovich & Koopmans.

Discrete OT is a linear program

source μ (top) and target ν (bottom)



cost c_{ij} (grey) and optimal plan γ_{ij} (blue)



$$\min_{\gamma \geq 0} \sum_{i,j} c_{ij} \gamma_{ij} \quad \text{s.t.} \quad \sum_j \gamma_{ij} = a_i, \quad \sum_i \gamma_{ij} = b_j.$$

Optimal plans sit on vertices of the transport polytope: at most $m + n - 1$ nonzero entries.

Kantorovich duality

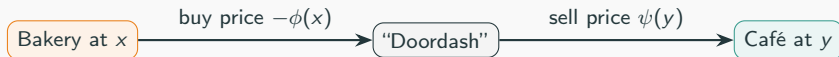
Kantorovich duality

Theorem (Kantorovich duality)

Let $c : \mathbb{R}^d \times \mathbb{R}^d \rightarrow [0, \infty]$ be lower semicontinuous. Then

$$\min_{\gamma \in \Pi(\mu, \nu)} \int c d\gamma = \sup \left\{ \int \phi d\mu + \int \psi d\nu : \phi(x) + \psi(y) \leq c(x, y) \quad \forall x, y \right\}.$$

The supremum may be restricted to pairs (ϕ, ϕ^c) , where $\phi^c(y) := \inf_x \{c(x, y) - \phi(x)\}$.

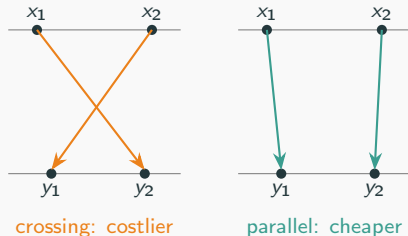


Prices are competitive if $\phi(x) + \psi(y) \leq c(x, y)$. Maximal revenue = minimal cost.

Villani, *Topics* (2003) Thm. 1.3; *Old & New* (2009) Thm. 5.10. Shipper story: Caffarelli.

Complementary slackness and cyclical monotonicity

- ▶ At optimality: $\phi(x) + \psi(y) = c(x, y)$ for γ -a.e. (x, y) .
- ▶ $\text{supp}(\gamma)$ is **c-cyclically monotone** $\Rightarrow$ **monotone** on the support of optimal plan,
$$c(x_i, y_i) + c(x_j, y_j) \leq c(x_i, y_j) + c(x_j, y_i).$$



$c(x, y) = |x - y|^2$ on $\mathbb{R}$:
optimal plans never cross.

Quadratic cost: the problem becomes about convexity

- ▶ Expand the cost: $\frac{1}{2} \|x - y\|^2 = \frac{1}{2} \|x\|^2 + \frac{1}{2} \|y\|^2 - x \cdot y$.
- ▶ The marginal terms are fixed, so

$$\text{minimise } \mathbb{E} \|Y - X\|^2 \iff \text{maximise the correlation } \mathbb{E}[X \cdot Y].$$

- ▶ Dual problem:

$$\sup_{\gamma \in \Pi(\mu, \nu)} \mathbb{E}_{\gamma}[X \cdot Y] = \inf_{\varphi \text{ convex}} \left\{ \int \varphi d\mu + \int \varphi^* d\nu \right\}.$$

Next: a short tour of convex analysis.

Convex functions and Brenier's theorem

Convex analysis: subdifferentials and gradients

- φ convex, lsc, proper. $y \in \partial\varphi(x)$ iff

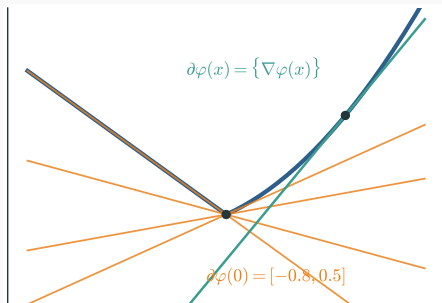
$$\varphi(z) \geq \varphi(x) + y \cdot (z - x) \quad \forall z.$$

- $\partial\varphi(x)$ is a convex set: slopes of supporting hyperplanes.

- **Monotone:**

$y_i \in \partial\varphi(x_i) \Rightarrow (y_2 - y_1) \cdot (x_2 - x_1) \geq 0$. In $d = 1$: increasing.

- $\partial\varphi(x) = \{y\}$ iff $y = \nabla\varphi(x)$.



Convex conjugates

- ▶ Legendre–Fenchel conjugate:

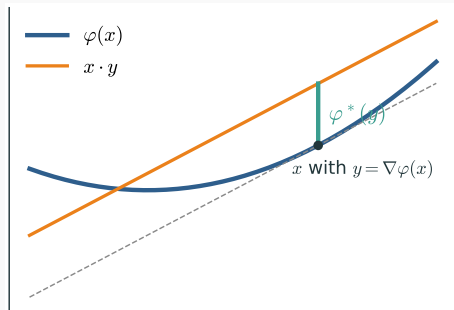
$$\varphi^*(y) = \sup_x \{x \cdot y - \varphi(x)\},$$

proper, lsc, convex.

- ▶ Fenchel–Young: $x \cdot y \leq \varphi(x) + \varphi^*(y)$.

- ▶ Equality case. TFAE:

1. $x \cdot y = \varphi(x) + \varphi^*(y)$;
2. $y \in \partial\varphi(x)$;
3. $x \in \partial\varphi^*(y)$.



$\varphi^*(y)$ is the largest gap between the line of slope y and φ .

Brenier's theorem (1991)

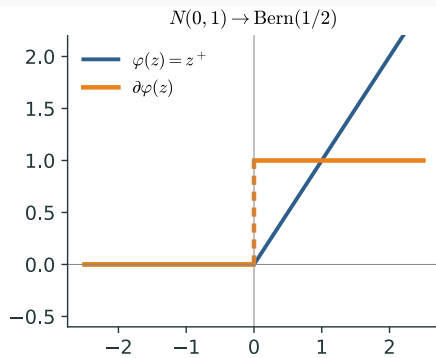
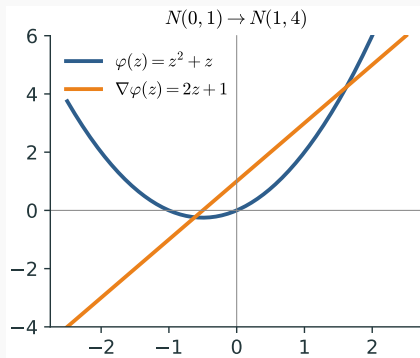
Let $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d)$, $c(x, y) = \|x - y\|^2$, and assume μ has a density. Then

- ▶ the Monge and Kantorovich problems have the **same unique** solution;
 - ▶ there is a convex $\varphi : \mathbb{R}^d \rightarrow \mathbb{R} \cup \{+\infty\}$ with $(\nabla\varphi)_\# \mu = \nu$;
 - ▶ $T = \nabla\varphi$ is the optimal map, unique μ -a.e.
-
- ▶ Conversely, *any* gradient of a convex function pushing μ to ν is optimal.
 - ▶ Monge maps = “monotone” maps; in $d = 1$, quantile-quantile map.

Brenier (1991) CPAM; McCann (1995) Duke.

Examples of Brenier's theorem

- ▶ $\mu = N(0, 1)$, $\nu = N(1, 4)$: $\nabla\varphi(z) = 2z + 1$, $\varphi(z) = z^2 + z$ convex. Optimal!
- ▶ $\mu = N(0, 1)$, $\nu = \text{Bern}(1/2)$: $T(z) = 1\{z > 0\} = \nabla\varphi(z)$, $\varphi(z) = z^+$.
 φ is not differentiable at 0, but $\mu(\{0\}) = 0$. No problem!



- ▶ Quadratic cost, $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d)$ arbitrary.
- ▶ **Knott–Smith:** $\pi \in \Pi(\mu, \nu)$ is optimal iff for some convex lsc φ

$$\text{supp}(\pi) \subseteq \text{Graph}(\partial\varphi), \quad \text{i.e.} \quad x \cdot y = \varphi(x) + \varphi^*(y) \quad \pi\text{-a.s.}$$

- ▶ This is complementary slackness from finite-dimensional linear programming.
- ▶ **Brenier:** if μ is absolutely continuous, $\partial\varphi$ is single-valued μ -a.e., so

$$\pi = (\text{Id} \times \nabla\varphi)_\# \mu \quad \text{is unique.}$$

Knott & Smith (1984) JOTA; Rüschendorf & Rachev (1990).

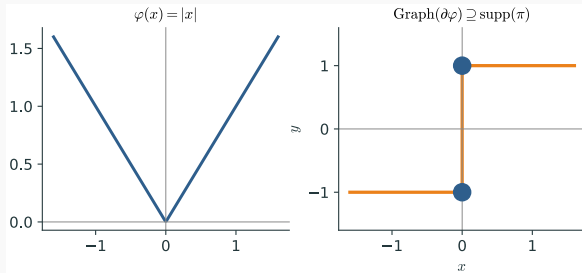
Knott–Smith holds even when Brenier fails

► $\mu = \delta_0$, $\nu = \frac{1}{2}\delta_1 + \frac{1}{2}\delta_{-1}$.

- The only coupling (hence optimal):

$$\pi(\{(0, 1)\}) = \pi(\{(0, -1)\}) = \frac{1}{2}.$$

- $\varphi(x) = |x|$: $\partial\varphi(0) = [-1, 1]$, so $\text{supp}(\pi) \subseteq \text{Graph}(\partial\varphi)$.
- No Brenier map: mass at 0 must split.



Beyond quadratic cost: the twist condition

- ▶ General cost $c(x, y)$, differentiable in x . **Twist condition:**

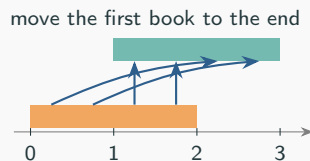
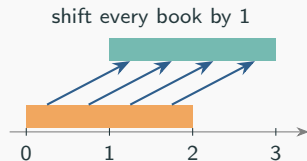
for each x , $y \mapsto \nabla_x c(x, y)$ is injective.

- ▶ **Theorem** (Gangbo–McCann 1996). $\mu \ll \text{Leb}$, c twisted and locally semiconcave, finite optimal cost $\Rightarrow$ unique optimal plan, induced by a map.
- ▶ $c = h(x - y)$, h strictly convex: $T(x) = x - (\nabla h)^{-1}(\nabla \phi(x))$. For $h = \frac{1}{2} |z|^2$ this is Brenier's $T = \nabla(\frac{1}{2} |x|^2 - \phi)$.

Gangbo & McCann (1996) Acta Math.; Villani (2009) Ch. 10.

When twist fails: book shifting

- ▶ $c(x, y) = |x - y|$ on $\mathbb{R}$ is not strictly convex: $\partial_x c = \text{sign}(x - y)$ is not injective in y .
- ▶ $\mu = \text{Unif}[0, 2]$, $\nu = \text{Unif}[1, 3]$. Two very different optimal maps:



	shift $x \mapsto x + 1$	block move
cost $ x - y $	1	$\frac{1}{2} \cdot 2 = 1$
cost $ x - y ^2$	1	$\frac{1}{2} \cdot 4 = 2$

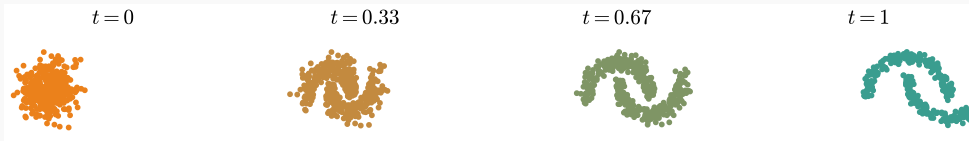
Non-uniqueness for $|x - y|$; the quadratic cost

picks the shift.

McCann interpolation
(quadratic cost only from now on)

McCann interpolation

- ▶ μ absolutely continuous, quadratic cost, Brenier map $T = \nabla\varphi$ from μ to ν .
- ▶ For $t \in [0, 1]$, $T_t(x) := (1 - t)x + tT(x)$.
- ▶ $X_0 \sim \mu$, let $X_t = T_t(X_0)$. $\rho_t = \text{law of } X_t$, is the marginal flow.
- ▶ $(\rho_t)_{0 \leq t \leq 1}$: **McCann** (displacement) interpolation from μ to ν .



McCann (1997) Adv. Math.

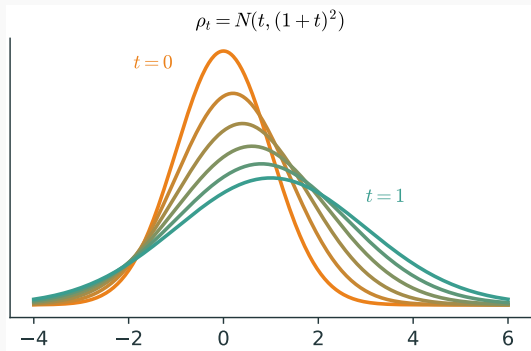
McCann interpolation: examples

Gaussians. $\mu = N(0, 1)$, $\nu = N(1, 4)$.

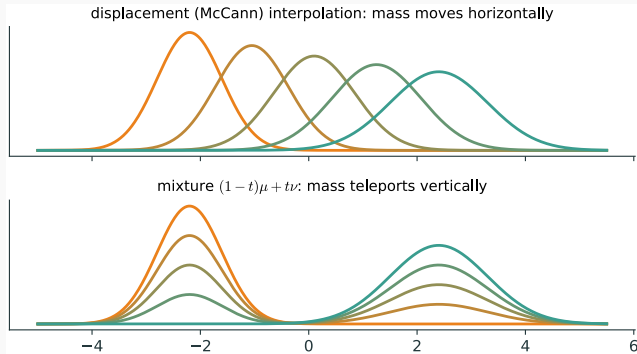
- ▶ $T(x) = 2x + 1$, so
 $T_t(x) = (1 + t)x + t$.
- ▶ $\rho_t = N(t, (1 + t)^2)$: normal shape preserved.

Cubes. $\mu = \text{Unif}([0, 1]^d)$,
 $\nu = \text{Unif}([0, r]^d)$.

- ▶ $T(x) = rx$, $T_t(x) = (1 - t + tr)x$.
- ▶ $\rho_t = \text{Unif}([0, 1 - t + tr]^d)$: shape preserved again.



Displacement versus mixture



- ▶ Mixture $(1-t)\mu + t\nu$: the *linear* structure of measures. Mass fades out and in.
- ▶ Displacement: the *geometry of $\mathbb{R}^d$* is lifted to measures. Mass travels.
- ▶ Different interpolations give different notions of convexity.

Dynamic optimal transport

$$\mathbb{W}_2^2(\mu, \nu) = \inf_{\gamma \in \Pi(\mu, \nu)} \int \|y - x\|^2 d\gamma$$

- ▶ Instead of static transport, optimise over flows $(T_t)_{0 \leq t \leq 1}$, $T_0 = \text{Id}$, $(T_1)_\# \mu = \nu$.
- ▶ $X \sim \mu$. $z_t := T_t(X)$.

$$\text{Kinetic energy} = \inf \mathbb{E} \left[\int_0^1 \|\dot{z}_t\|^2 dt \right].$$

- ▶ Which a.e. differentiable flow transporting μ to ν has minimal kinetic energy?
- ▶ Optimization over stochastic processes with smooth paths.

McCann interpolations are geodesics

1. By Jensen (or Cauchy–Schwarz), for any path with endpoints z_0, z_1 :

$$\int_0^1 \|\dot{z}_t\|^2 dt \geq \left\| \int_0^1 \dot{z}_t dt \right\|^2 = \|z_1 - z_0\|^2,$$

with equality iff $z_t = (1 - t)z_0 + tz_1$ (straight line, constant speed).

2. Hence the dynamic problem reduces to choosing endpoints $z_1 = T_1(x)$: this is Monge's problem, solved by Brenier's T .
3. Optimal flow: $T_t(x) = (1 - t)x + tT(x)$, i.e. **the McCann interpolation**.

McCann interpolation are Constant-speed geodesic

$$\mathbb{W}_2(\rho_t, \rho_s) = |t - s| \mathbb{W}_2(\mu, \nu), \quad 0 \leq s, t \leq 1.$$

Eulerian vs. Lagrangian viewpoints

- ▶ Suppose we are given a velocity field at each time-point $v_t : \mathbb{R}^d \rightarrow \mathbb{R}^d$.
- ▶ $X_0 \sim \mu$. Cauchy problem:

$$\frac{d}{dt}X_t = v_t(X_t), \quad t \in (0, 1).$$

- ▶ ρ_t = density of X_t satisfies the **continuity equation**

$$\partial_t \rho_t + \nabla \cdot (\rho_t v_t) = 0, \quad \rho_0 = \mu.$$

- ▶ Tracking the path X_t is called Lagrangian viewpoint.

Benamou–Brenier: the Eulerian picture

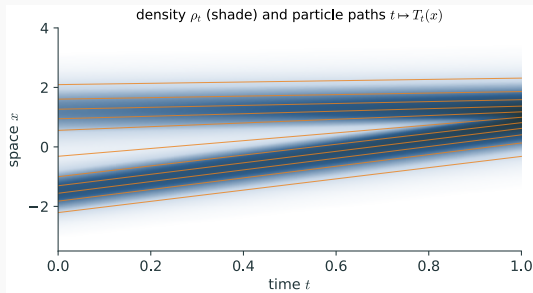
- ▶ Track densities ρ_t and velocity fields v_t instead of particles (Eulerian viewpoint):

$$\partial_t \rho_t + \nabla \cdot (\rho_t v_t) = 0, \quad \rho_0 = \mu, \quad \rho_1 = \nu.$$

- ▶ **Theorem** (Benamou–Brenier 2000):

$$\mathbb{W}_2^2(\mu, \nu) = \inf_{(\rho, v)} \int_0^1 \int \|v_t\|^2 d\rho_t dt.$$

- ▶ Optimiser: McCann interpolation with $v_t = (T - \text{Id}) \circ T_t^{-1}$.



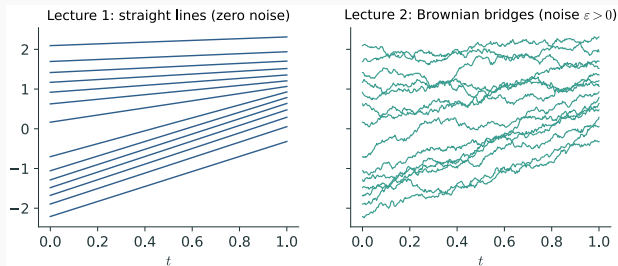
μ bimodal $\rightarrow \nu$ Gaussian on $\mathbb{R}$; paths never cross.

Dynamic OT on the path space

- ▶ Let P be a law on paths $C([0, 1]; \mathbb{R}^d)$ with $X_0 \sim \mu$, $X_1 \sim \nu$:

$$\mathbb{W}_2^2(\mu, \nu) = \min_P \mathbb{E}_P \int_0^1 \|\dot{X}_t\|^2 dt.$$

- ▶ Optimal P : law of $t \mapsto (1 - t)X + tT(X)$, $X \sim \mu$. Random initial point, deterministic straight path.



Next lecture: kinetic energy $\rightsquigarrow$ relative entropy w.r.t. Brownian motion: Schrödinger bridges.

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See you next time!

Lecture 2: Entropy-regularised OT and Schrödinger bridges